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Mesozoic paleogeography of the Northern Calcareous Alps—Evidence from paleomagnetism and facies analysis

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ABSTRACT

The Northern Calcareous Alps are generally thought to have been part of the same southern Tethyan continental margin as the southern Alps during Late Jurassic and Early Cretaceous time. Contrasting Liassic paleomagnetic data from the Northern Calcareous Alps and the southern Alps can be explained by large-scale clockwise rotation during Late Cretaceous and Tertiary tectonic emplacement of the Northern Calcareous Alps onto the European continental margin. Alternatively, the data can be reconciled by postulating that the Northern Calcareous Alps did not undergo the African Mesozoic rotational history seen in the southern Alps and were separated from the southern Alps by a short-lived branch of Jurassic ocean.

Facies analysis in the Karwendel-Thiersee syncline of the Northern Calcareous Alps indicates that the Jurassic basin sediments in this area were deposited in an east-west–oriented sinistral strike-slip fault system. In the southern Alps and central Austroalpine complex, coeval basin sedimentation was controlled by north-south–oriented normal faults. This contrasting basin evolution and evidence for a source of ophiolitic detritus south of the Northern Calcareous Alps argue for a short-lived ocean basin south of the Northern Calcareous Alps that was closed by Early Cretaceous time.

INTRODUCTION

The Northern Calcareous Alps comprise a belt of thrust sheets composed mainly of Middle and Upper Triassic shallow-water carbonate rocks. The belt extends from close to the Swiss-Austrian border to the Vienna basin (Fig. 1), a distance of about 500 km. The average width of the belt is about 50 km. These nappes were thrust over the flysch and molasse sedimentary rocks associated with the northern Tethyan (European) continental margin and are believed to have been derived from south of the Penninic ocean and, therefore, to have been part of the southern Tethyan continental margin (e.g., Bosellini et al., 1980; Weissert and Bernoulli, 1985; Tollman, 1987a). Here we present data that may contradict this generally assumed paleogeography.

The stratigraphy of the Northern Calcareous Alps can be subdivided into three stages: the continental redbed and siliciclastic shallow-water stage (Late Permian and Scythian), the shallow-water carbonate and short-lived basinal stage (Middle and Late Triassic), and the deeper water carbonate stage (Jurassic) (Schlager and Schollnberger, 1975; Lein, 1987). Although largely eroded, isolated outcrops of Jurassic deeper-water deposits are widely distributed throughout the belt; however, exposure is often limited to the Liassic part of the sequence. The paleomagnetic data presented here are derived from the Adnet limestones of Liassic (Hettangian to Toarcian) age, which overlie Rhaetian shallow-water carbonate rocks. The Adnet Limestone is a fine-grained, red, marly, sometimes nodular limestone deposited in basins or on submarine swells.

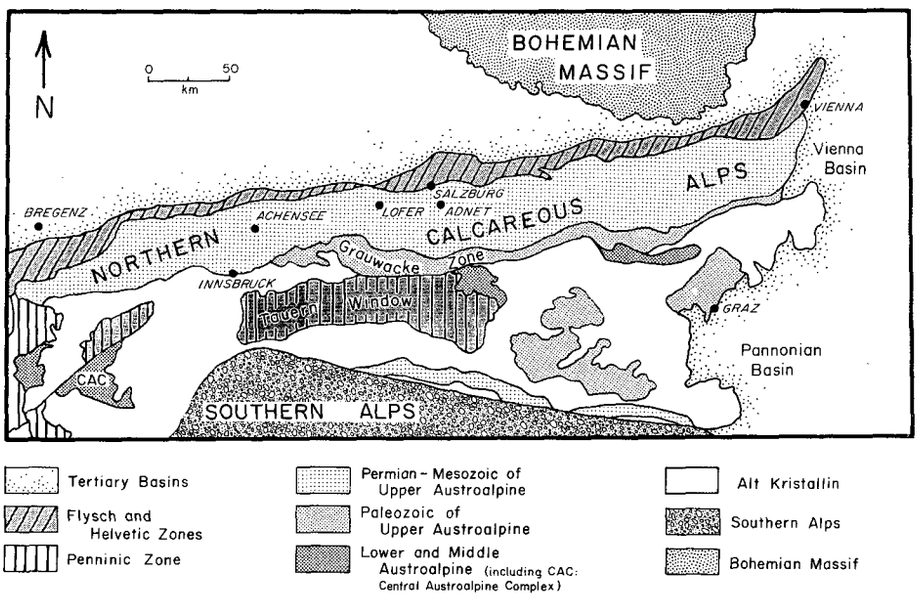


Figure 1. Simplified structural map of Austria.

PALEOMAGNETIC DATA

New paleomagnetic data from 24 sites of Adnet Limestone from three regions of the Northern Calcareous Alps (Achensee, Lofer, and Adnet areas) (Fig. 1) yield high blocking-temperature magnetization components (Fig. 2); most of the sites exhibit dual polarity. High blocking-temperature magnetization components were computed from orthogonal projections of thermal demagnetization data using principal component analysis (Kirschvink, 1980). Site mean directions for the Karwendel-Thiersee syncline at Achensee pass the fold test at the 99% confidence level (McElhinny, 1964), indicating a pre-Tertiary or pre-Late Cretaceous age for the high blocking-temperature component (Fig. 3, Table 1).

Whereas the mean inclinations from the Northern Calcareous Alps are consistent with values expected for both the southern Alps and for Europe, the mean declinations are significantly different from one another, and from values that would fit the African or Eurasian apparent polar wander paths (APWP) (Table 1). The mean direction obtained by Mauritsch and Frisch (1978) from Jurassic limestones in the Osterhorn region east of Salzburg is included in Table 1 (see also Mauritsch and Becke, 1987). For comparison of these data with those from stable Europe, we use the Early Jurassic (190 Ma) point on the Eurasian APWP of Westphal et al. (1986), which was derived by rotating the North American data according to the North

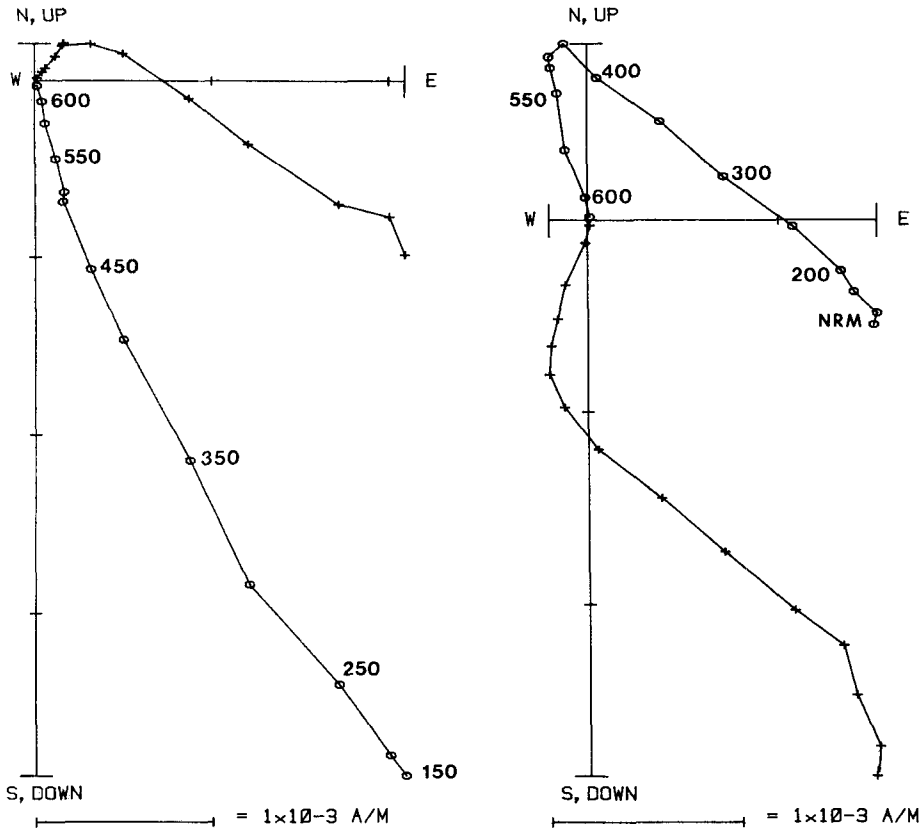


Figure 2. Orthogonal projections of thermal demagnetization data for two samples from Karwendel/Thiersee syncline, after correction for structural tilt. Crosses and open symbols represent projection on horizontal and vertical planes, respectively. Demagnetization temperatures for certain points are given in °C. NRM is natural remanent magnetization.

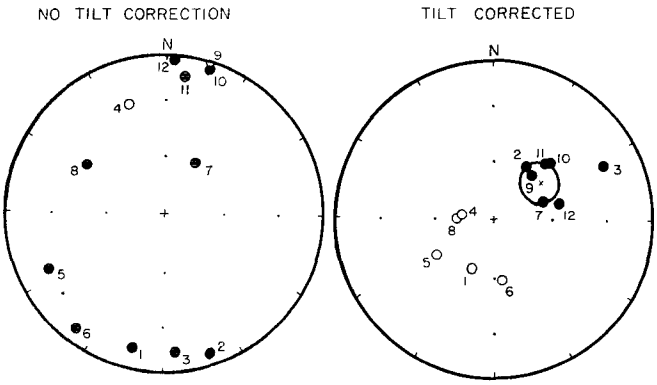


Figure 3. Equal-area projection of site mean characteristic magnetization directions from Karwendel-Thiersee syncline, before (left) and after (right) structural tilt correction. Solid and open symbols represent positive (downward) and negative (upward) inclinations, respectively. Circle of 95% confidence about mean is shown.

TABLE 1. JURASSIC PALEOMAGNETIC DATA FROM THE NORTHERN CALCAREOUS ALPS

Locality	Sites	Before tilt correction Dec. (o)	Inc. (o)	k	a95	After tilt correction Dec. (o)	Inc. (o)	k	a95
Karwendel/ Thiersee	12	50.1	8.6	1.6	53.3	50.5	59.7	19.3	10.1
Lofer	8	357.6	52.4	31.4	10.0	15.1	55.0	71.2	6.6
Adnet	3	79.4	47.5	149.2	10.1	80.8	57.1	63.9	15.5
Osterhorn*	8					62.0	45.0	29.1	11.3
Direction in the Karwendel-Thiersee to fit African APWP						335.0	52.0		
				Eurasian APWP		24.0	58.0		

Note: Dec.= declination, Inc.= inclination, k and a95 correspond to precision and the radius of the circle of 95% confidence from Fisher (1953) statistics. APWP is apparent polar wander path.
* Mauritsch and Frisch (1978).

Atlantic reconstruction of Savostin et al. (1986). This path is consistent with Middle Jurassic paleomagnetic data from Poland (Kadzialko-Hofmokl and Kruczyk, 1987). For comparison with the southern Alps, we use the Early Jurassic (190 Ma) African pole of Westphal et al. (1986), again derived by rotating the North American data. The lack of Early Jurassic paleomagnetic data from the southern Alps does not permit a direct comparison between magnetization directions from the Northern Calcareous Alps and the southern Alps. The use of the African path as a proxy for southern Alps data is based on the observed consistency of Late Jurassic and Cretaceous southern Alps data with the African path (VandenBerg and Wonders, 1976; Lowrie et al., 1980; Lowrie, 1986; Channell, 1986).

FACIES ANALYSIS

The isolated outcrops of Jurassic basin sedimentary rocks in the Northern Calcareous Alps do not generally provide a sufficiently detailed facies record to elucidate Jurassic basin geometry. However, the Karwendel-Thiersee syncline, located close to Achensee, an alpine lake 40 km northeast of Innsbruck (Fig. 1), provides a detailed record of Jurassic basin sedimentation in the area. The close to tight syncline is about 60 km long and is characterized by two segments with east-west fold axes joined by a more complexly deformed northeast-oriented segment (Fig. 4). The syncline includes Upper Triassic shallow-water limestones and dolomites overlain by a sequence of Liassic to Lower Cretaceous deeper-water sedimentary rocks (Spieler, 1990). Facies analysis indicates late Hettangian to Sinemurian extensional tectonism with the formation of a complicated rift-basin topography, because Rhaetian shallow-water deposits are abruptly overlain by a gray cyclic limestone-marl sequence in the deeper parts of the basin system (Lower Allgau Formation), condensed red limestones on the submarine swells (Adnet Formation), and gray, spicule-rich limestones on the basin slopes (Scheibelberg Formation). The rate of extension slowed during the early Pliensbachian and increased during the Pliensbachian-Toarcian; there are frequent scarp breccias and mass flows in the basin deposits. The lower Toarcian in the deeper parts of the basin system is characterized by the formation of bituminous marls (Middle Allgau Formation) coincident with a global anoxic event (Jenkyns, 1988). Red marly limestone deposition continued on the basin swells and submarine margins at this time. Gray marl-limestone cycles (Upper Allgau Formation) developed in the late Toarcian. Further subsidence of the basin in Callovian time resulted in the deposition of radiolarites in the deeper parts of the basin system (Ruhpoldinger Formation), followed by Malm aptychus limestones, and finally Early Cretaceous "Neocomian" marls.

The facies distribution of the surrounding upper Rhaetian shallow-water sedimentary rocks, as well as Jurassic slope and basin facies zoning, indicate deepening water depth toward the axis of syncline and toward the central northeast-trending part of the syncline system (Spieler, 1990), particularly at times of accelerated subsidence in the late Toarcian and Callovian-Oxfordian. The facies analysis is consistent with the hypothesis that the Jurassic sediments were deposited in a sinistral strike-slip fault zone, the central northeast-striking part of the basin system being a fossil pull-apart basin. The above interpretation assumes that the S shape of the Karwendel-Thiersee syncline is not due to superimposed Alpine deformation phases. The general consistency in paleomagnetic declination (Fig. 4), implying a lack of differential rotation (except for site 6), is consistent with this interpretation. Although Jurassic strike-slip faults are not directly observed, we postulate that the *Alpine* strike-slip faults that bound Triassic sedimentary rocks that are thrust over the central part of the basin system from the east (Fig. 4) are reactivated Jurassic strike-slip faults. The present east-west strike of the hypothetical Jurassic sinistral strike-slip fault zone is kinematically consistent with the northeast-southwest orientation of Liassic neptunian dikes in Triassic shallow-water carbonates exposed in the Rofan area, 5 km east of Achensee (Fig. 4).

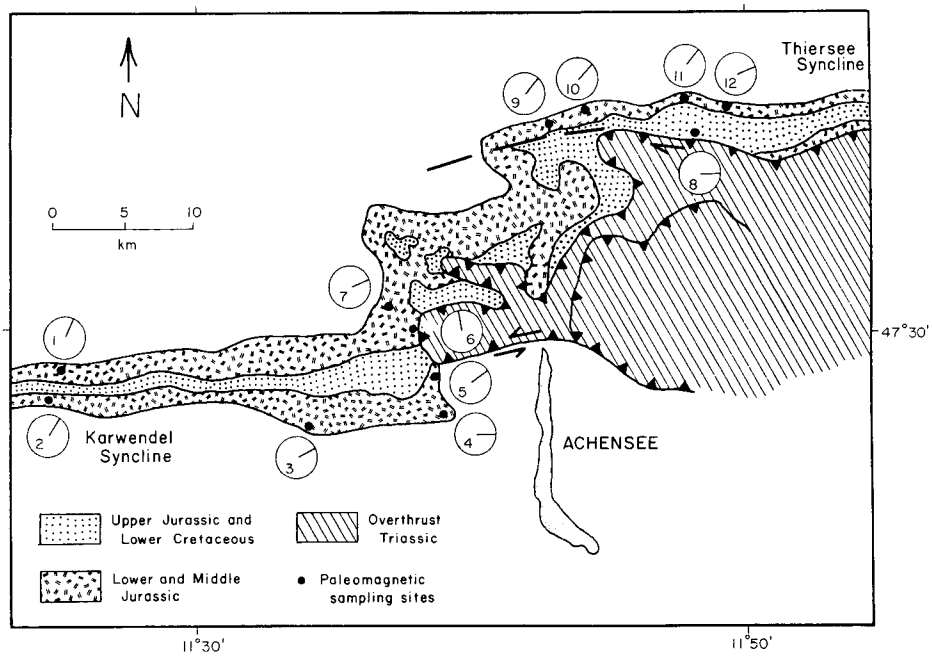


Figure 4. Sketch of part of Karwendel-Thiersee syncline indicating outcrops of Jurassic and Cretaceous sedimentary rocks and paleomagnetic sampling sites. Site mean declinations are diagramatically represented.

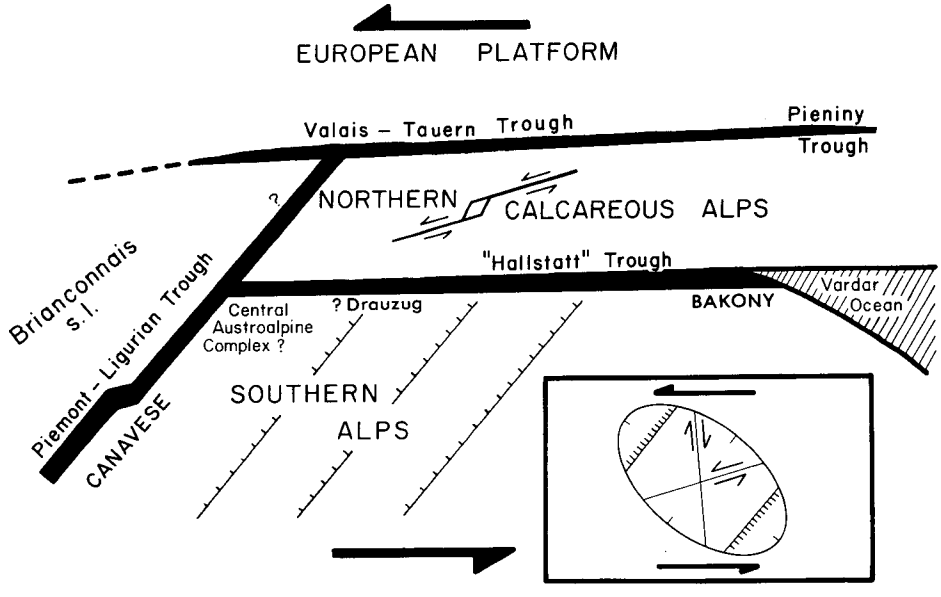


Figure 5. Schematic Liassic paleogeography relative to lines of later (Middle Jurassic) oceanic rifting. Today's north-south-oriented paleotectonic normal faults in southern Alps and central Austroalpine complex were oriented approximately N50°E in Liassic time, relative to present-day Europe. Liassic orientation of inferred Karwendel-Thiersee sinistral strike-slip fault is shown.

The control of sedimentation along east-west–oriented strike-slip faults, inferred from the data from the Karwendel-Thiersee syncline, contrasts with the coeval half-graben basin geometry found in the southern Alps, the central Austroalpine complex, the Drauzug, and the Bakony, where Jurassic sedimentation was controlled by north-south–oriented normal faults (Winterer and Bosellini, 1981; Eberli, 1988; Blau and Schmidt, 1988; Kazmer and Kovacs, 1985).

DISCUSSION

It is generally believed that the early opening of the Jurassic Tethys in this region occurred in an east-west–oriented sinistral transtensional kinematic framework (see Weissert and Bernoulli, 1985; Savostin et al., 1986; Lemoine and Trumpy, 1987; Trumpy, 1988). The orientation of strike-slip faults within this framework can be predicted from wrench tectonic models (Wilcox et al., 1973). Synthetic sinistral strike-slip faults making an angle of 10°–30° to the wrench strike, and antithetic dextral strike-slip faults making an angle of 70°–90° to the wrench strike are predicted (Fig. 5). The maximum extension direction would approximately bisect the obtuse angle between the strike-slip faults. In the southern Alps, the present orientation of Jurassic paleotectonic normal faults is north-south (Bosellini et al., 1980; Winterer and Bosellini, 1981). A similar north-south orientation of paleotectonic normal faults has been recognized in the central Austroalpine complex of eastern Switzerland (Eberli, 1988; Froitzheim, 1988). In Liassic time, prior to oceanic rifting and the Mesozoic African rotation of the southern Alps, these faults would have been oriented N50E relative to present Europe (Fig. 5). The inferred sinistral strike-slip fault zone in the Northern Calcareous Alps would have been oriented approximately N65E, relative to present Europe, in Liassic time (Fig. 5). The orientation and style of faulting in the Northern Calcareous Alps and southern Alps are therefore consistent with the prediction of wrench tectonic models (Fig. 5).

If the Northern Calcareous Alps were part of the southern Alpine continental margin prior to tectonic emplacement, clockwise rotations of 75°, 40°, 105°, and 87° are required during tectonic emplacement for the four localities listed in Table 1. If, however, the Northern Calcareous

Alps did not undergo the African rotation seen in the southern Alps, but were unrotated relative to Europe prior to tectonic emplacement, then the clockwise rotations of 26°, -10° (anticlockwise), 57°, and 38° during tectonic emplacement are required. The latter alternative, involving smaller amounts of rotation, is more acceptable in view of the large scale of the individual thrust sheets in the Northern Calcareous Alps, and implies the existence of a branch of Mesozoic Tethys south of the Northern Calcareous Alps.

There are further lines of evidence that favor a Jurassic mobile zone between the southern Alps and the Northern Calcareous Alps. From the region of Salzburg eastward, widespread Late Jurassic thrusting affected the Juvavikum nappes of the Northern Calcareous Alps. Upper Triassic basin sedimentary rocks (Halstatt facies) were thrust over Upper Jurassic basin sedimentary rocks during Oxfordian to Early Cretaceous time (Tollmann, 1987b). According to Decker et al. (1987) and Pober and Faupl (1988), ophiolitic detritus in Lower Cretaceous flysch deposits in the Northern Calcareous Alps indicate an eroding ophiolitic body to the (present-day) south of the Northern Calcareous Alps. Ophiolitic detritus with southern provenance is observed in Upper Cretaceous and Paleogene Gosau sedimentary rocks, although in these deposits there is a more dominant ophiolitic detrital source to the north. Winkler (1988) explained ophiolitic detritus in middle and Upper Cretaceous flysch of the western Northern Calcareous Alps in terms of southward obduction and erosion of ophiolitic nappes. An ophiolitic source to the south of the Northern Calcareous Alps would provide an alternative to the obduction hypothesis. Widespread anchimetamorphism, particularly in the southern part of the Northern Calcareous Alps, is dated close to the time of the Jurassic/Cretaceous boundary (Kralik et al., 1987).

These data argue for a zone of mobility, partly floored by oceanic crust, separating the Northern Calcareous Alps and southern Alps during at least part of Jurassic time. This scenario leads to a paleogeography in which the Northern Calcareous Alps occupy an intermediate position in the Jurassic Tethys, analogous to that of the Brianconnais (Fig. 5). The region of extended (oceanic) crust between the Northern Calcareous Alps and the southern Alps is likely to have closed by Early Cretaceous time.

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